THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 5: **Describing Motion Around Us** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** scalar	Magnitude only.
2	**(c)** displacement	Area.
3	**(d)** 50 km/h	100/2.
4	**(a)** vector	Magnitude and direction.
5	**(a)** acceleration	Slope.
6	**(b)** 2 m/s	10/5.
7	**(d)** distance/time	Formula.
8	**(c)** zero	Back at start.
9	**(c)** 5 m/s squared	10/2.
10	**(d)** displacement/time	Formula.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Total length of the path travelled (2).	2
12	Rate of change of velocity (2).	2
13	s = ut + (1/2)at squared (1); Meaning of symbols (1).	1+1
14	Shortest distance between initial and final positions with direction (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Scalar and vector (1); Distance is always positive (1); Displacement may be zero (1).	1+1+1
16	108 x 5/18 (1); = 30 (1); m/s (1).	1+1+1
17	Scalars: distance, speed, time (1); Vectors: displacement, velocity, acceleration (1); Direction is the difference (1).	1+1+1
	OR: Scalar vs vector (1); Distance vs displacement (1); Speed is never negative (1).	1+1+1
18	20 x 18/5 (1); = 72 (1); km/h (1).	1+1+1
	OR: Pythagoras (1); sqrt(16 + 9) (1); 5 m (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Distance (1); Displacement (1); Speed (1); Velocity (1); Acceleration (1).	1+1+1+1+1
	OR: Total distance 100 km (1); Total time 3 h (1); 33.3 km/h (1); Formula (1); Units (1).	1+1+1+1+1
20	a = (v - u)/t (1); at = v - u (1); v = u + at (1); Symbols (1); Conditions (1).	1+1+1+1+1
	OR: t = 5 s (1); s = 400/8 = 50 m (1); v = 0 (1); Formula (1); Units (1).	1+1+1+1+1

Mistakes to avoid:

- Using distance instead of displacement (or speed instead of velocity); for a round trip the displacement and average velocity are zero.
- Not converting km/h to m/s before using the equations of motion, or mixing km with seconds.
- Putting a positive sign for retardation in $v = u + at$ or $v^2 = u^2 + 2as$; for a slowing body a is negative.
- Reading the slope of an x-t graph as acceleration and the slope of a v-t graph as velocity; slope of x-t is velocity, slope of v-t is acceleration.
- Calculating the distance from a v-t graph by adding signed areas or by taking only one segment; add the areas of all segments (triangles and rectangles) and take area below the axis as positive for distance.
- Thinking that acceleration is zero at the top of a vertical throw; only the velocity is zero, acceleration is still g.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 5: **Describing Motion Around Us** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 5/18 m/s	1000/3600.
2	**(d)** 100 m	20 x 5.
3	**(a)** change in velocity/time	Formula.
4	**(c)** 10	36 x 5/18.
5	**(d)** 10 m/s	gt.
6	**(a)** m/s	Metre per second.
7	**(a)** 5	18 x 5/18.
8	**(c)** 20 m	0.5 x 10 x 4.
9	**(a)** m/s squared	Unit.
10	**(c)** 10 m/s squared	g.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Equal distances in equal intervals of time (2).	2
12	v squared = u squared + 2as (1); Meaning of symbols (1).	1+1
13	Distance travelled per unit time (2).	2
14	54 x 5/18 (1); = 15 m/s (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Speed = distance/time (1); 150/3 (1); 50 km/h (1).	1+1+1
16	v = gt (1); = 20 (1); m/s (1).	1+1+1
17	Car in traffic (1); Falling ball (1); Train starting (1).	1+1+1
	OR: a = (v - u)/t (1); 20/5 (1); 4 m/s squared (1).	1+1+1
18	(a) Horizontal line (1); (b) Straight sloping line (1); (c) Curve bending upward (1).	1+1+1
	OR: Speedometer: instantaneous speed (1); Odometer: distance covered (1); Units km/h and km (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	a = 2 m/s squared (1); s = 0.5 x 2 x 100 (1); = 100 m (1); Formulas used (1); Units (1).	1+1+1+1+1
	OR: Reference point (1); Passenger in train (1); Platform observer (1); Trees seem to move (1); Motion is relative (1).	1+1+1+1+1
20	45 = 5 t squared (1); t = 3 s (1); v = 30 m/s (1); Starts from rest (1); Formulas (1).	1+1+1+1+1
	OR: Slope of s-t (1); Take two points (1); Slope of v-t (1); Rise by run (1); Units (1).	1+1+1+1+1

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 5: **Describing Motion Around Us** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 6 m/s	2 x 3.
2	**(a)** zero	No motion.
3	**(b)** zero	Dropped.
4	**(d)** negative acceleration	Slowing.
5	**(d)** zero	No change in velocity.
6	**(c)** the highest point	Turning point.
7	**(b)** a horizontal line	No change.
8	**(b)** speed	Slope.
9	**(d)** changing	Direction changes.
10	**(b)** a horizontal line	No change.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Acceleration due to gravity (1); About 10 m/s squared (1).	1+1
12	Displacement per unit time (2).	2
13	v = u + at (1); Meaning of symbols (1).	1+1
14	Zero (1); Initial and final points coincide (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	v = u + at (1); 10 + 10 (1); 20 m/s (1).	1+1+1
16	(a) Horizontal line (1); (b) Rising straight line (1); (c) Falling straight line (1).	1+1+1
17	Distance = 100 m (1); 100/5 (1); 20 m/s (1).	1+1+1
	OR: s = (1/2)at squared (1); 0.5 x 2 x 25 (1); 25 m (1).	1+1+1
18	Fan blades (1); Earth around the Sun (1); Second hand of a clock (1).	1+1+1
	OR: m (1); m/s (1); m/s squared (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	s-t graphs (1); v-t graphs (1); Slope meaning (1); Area meaning (1); Examples (1).	1+1+1+1+1
	OR: Speed = 4 m/s (1); Displacement = 0 (1); Distance 400 m (1); Velocity zero (1); Reason (1).	1+1+1+1+1
20	Definition (1); Constant speed (1); Direction changes (1); Accelerated (1); Examples (1).	1+1+1+1+1
	OR: Magnitude (1); Direction (1); Examples (1); Addition (1); Units (1).	1+1+1+1+1

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